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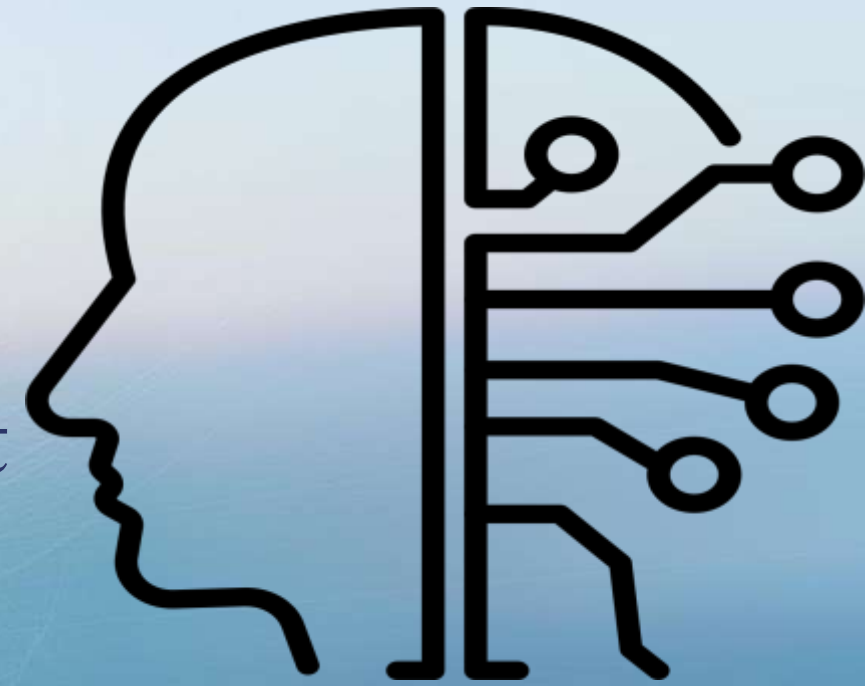


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992. Enhanced Network Attack Detection Using a TCN - BiLSTM–Transformer Framework

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Abstract

- Rapid growth of cyberattacks has increased the need for intelligent intrusion detection systems
- Traditional machine learning methods struggle with complex and low-frequency attacks
- This work proposes a hybrid deep learning model combining TCN, BiLSTM, and Transformer
- The model effectively captures temporal dependencies and global contextual relationships
- It improves detection accuracy for both common and rare attack patterns
- Performance is evaluated using metrics such as Accuracy, Precision, and F1-score
- Comparative analysis shows the proposed model outperforms Random Forest, CNN, CNN_x0002_LSTM, and CNN-BiLSTM
- The system is suitable for real-time and large-scale network security applications
- Proposed a hybrid model using TCN, BiLSTM, and Transformer
- Incorporates advanced preprocessing techniques such as feature selection and class balancing to enhance model performance
- Demonstrates strong generalization capability across benchmark datasets like NSL-KDD and UNSW-NB15
- Reduces false positives and false negatives, improving overall reliability of intrusion detection systems

Contents

- Introduction
- Literature Survey
- Objective
- Methodology
- Results/discussion
- Conclusion
- Future Scope
- References

Introduction

- Rapid growth of network technologies has led to a significant rise in sophisticated cyberattacks.
- Traditional Intrusion Detection Systems (IDS) fail to effectively detect complex and low-frequency attacks.
- Modern network traffic is highly dynamic, large-scale, and imbalanced, making detection more challenging.
- There is a critical need for intelligent systems that can learn temporal patterns and identify hidden threats.
- This project proposes a hybrid deep learning model combining TCN, BiLSTM, and Transformer architectures.
- The model captures both short-term and long-term dependencies in network traffic data.
- It improves detection accuracy, precision, and overall performance compared to existing models.
- The novelty lies in integrating temporal convolution with attention mechanisms for enhanced intrusion detection.
- Existing intrusion detection approaches lack the ability to understand sequential behavior and contextual relationships in network traffic
- Detection of multi-stage and stealthy attacks remains a major challenge in current security systems
- There is a need for a unified framework that integrates temporal learning and global attention for more accurate threat detection

Literature Survey

- Random Forest Based Intrusion Detection Systems highlights the effectiveness of traditional machine learning techniques for baseline intrusion detection.
- CNN-Based Intrusion Detection for Network Security demonstrates automatic feature extraction and improved detection performance using deep learning.
- CNN–LSTM Hybrid Model for Network Intrusion Detection shows enhanced capability in capturing spatial and temporal features of network traffic.
- BiLSTM-Based Intrusion Detection Systems improves detection accuracy by learning bidirectional dependencies.
- Transformer-Based Models for Cybersecurity Applications introduces attention mechanisms for handling complex and large-scale data patterns.
- Temporal Convolutional Networks for Sequence Modeling provides efficient temporal feature extraction with lower computational complexity.
- The proposed work builds upon these approaches by integrating TCN, BiLSTM, and Transformer for improved intrusion detection performance.

Objectives

- Develop an efficient and reliable network intrusion detection system
- Overcome limitations of traditional models in detecting complex and rare attacks
- Capture multi-level temporal dependencies in network traffic
- Handle class imbalance to improve detection of low-frequency attack types
- Reduce data complexity using effective feature selection techniques (BRFE)
- Integrate TCN, BiLSTM, and Transformer for enhanced feature learning
- Improve overall performance in terms of accuracy, precision, and F1-score
- Ensure scalability and adaptability for real-world network environments
- Enhance detection of multi-stage and stealthy attacks by modeling sequential network behavior over time
- Design a robust and generalized model that maintains consistent performance across different datasets and network conditions

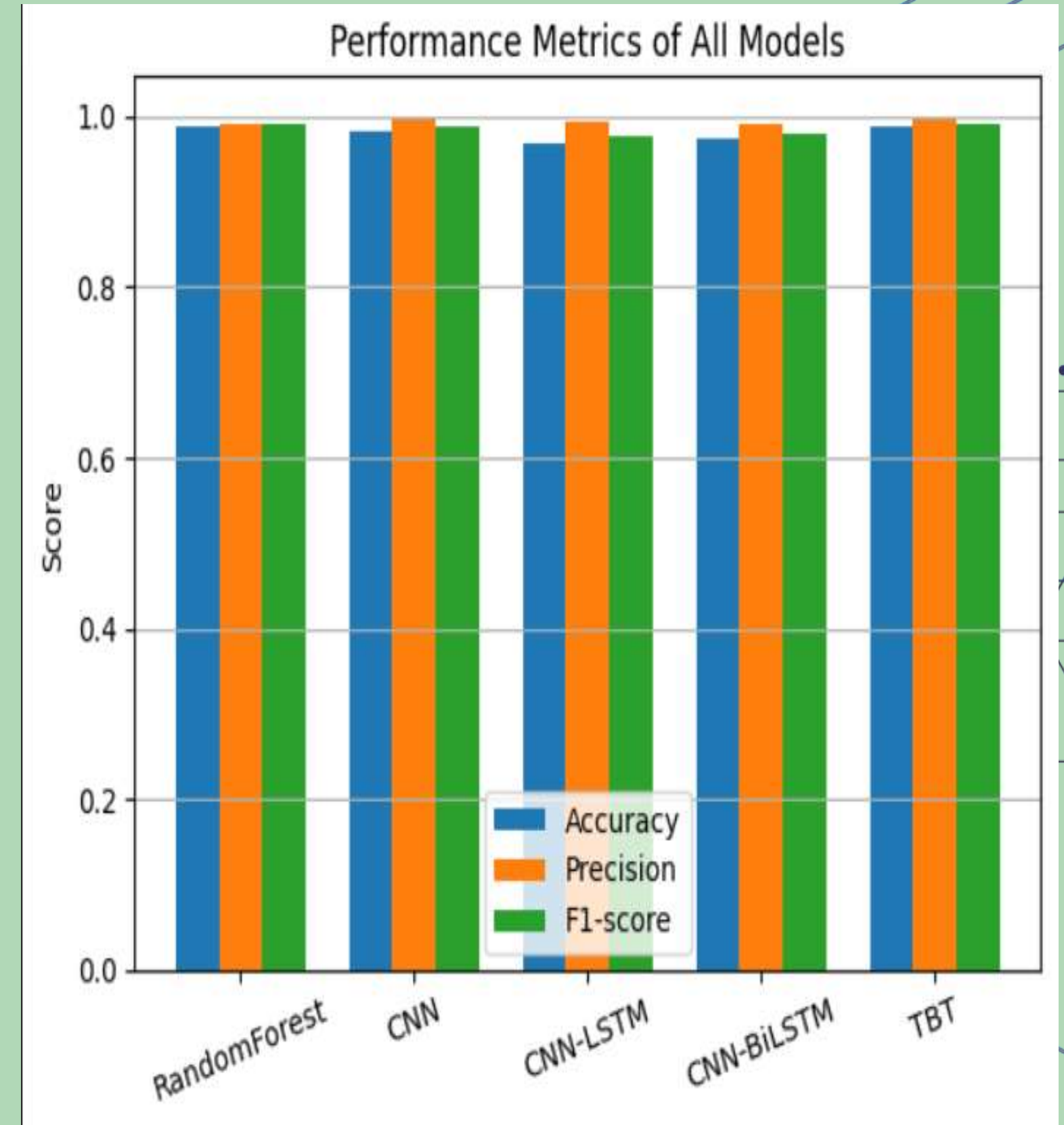
Methodology

- The system processes network traffic through preprocessing and feature extraction to prepare clean and structured input data
- TCN and BiLSTM modules capture both short-term and long-term temporal patterns in network behavior
- Feature fusion combines outputs from multiple models to improve representation and learning capability. Transformer encoder applies attention mechanism to capture global dependencies, leading to accurate classification of normal and attack traffic.



Results/discussion

- Proposed TCN–BiLSTM–Transformer (TBT) model achieves the highest performance across all metrics.
- Significant improvement observed compared to traditional (Random Forest) and baseline deep learning models.
- High precision and F1-score indicate effective detection of both normal and attack traffic.
- Consistent performance across metrics demonstrates strong model reliability and generalization.
- Hybrid architecture improves feature representation by combining temporal learning and attention mechanisms.
- Reduced false positives and false negatives enhance overall detection accuracy and system reliability



Conclusion

- Proposed hybrid TCN–BiLSTM–Transformer model provides effective intrusion detection
- Successfully captures short-term, long-term, and global dependencies in network traffic
- Achieves superior performance compared to traditional and deep learning baseline models
- Obtained highest accuracy of 98.6% with improved precision and F1-score
- Demonstrates strong ability to detect both common and rare cyberattacks
- Shows good generalization and stability across different datasets
- Suitable for real-time deployment in modern network security systems
- Demonstrates the effectiveness of hybrid deep learning by combining multiple architectures for improved intrusion detection performance
- Efficient handling of class imbalance improves detection of low-frequency yet critical attack types
- Robust preprocessing and feature selection techniques enhance model accuracy and reduce data complexity

Future Scope

- Extend the system to multi-class classification for identifying different types of attacks such as DoS, Probe, R2L, and U2R
- Implement real-time intrusion detection for faster response in live network environments
- Optimize the model for edge deployment on IoT devices and network gateways
- Integrate Explainable AI (XAI) techniques to improve transparency and interpretability of model decisions
- Use federated learning for secure and collaborative model training across multiple organizations
- Enhance detection of evolving and stealthy attacks using advanced temporal behavior modeling
- Incorporate adaptive learning mechanisms to continuously update the model with new attack patterns and improve long-term detection accuracy

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